

OAC Dublin common bean

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Smith, T. H., Michaels, T. E. and Pauls, K. P. 2010. **OAC Dublin common bean**. Can. J. Plant Sci. **90**: 511–514. OAC Dublin (CFIA registration no. 6599) is an indeterminate bush white bean (*Phaseolus vulgaris* L.) cultivar intended for use in areas with greater than 2600 crop heat units (CHU). It has excellent yield potential in either wide or narrow row production. Its seed has acceptable cooking and canning quality.

Key words: *Phaseolus vulgaris* L., white bean, common bean, cultivar description

Smith, T. H., Michaels, T. E. et Pauls, K. P. 2010. **Le haricot OAC Dublin**. Can. J. Plant Sci. **90**: 511–514. OAC Dublin est un cultivar nain de haricot blanc (*Phaseolus vulgaris* L.) à type de croissance indéfinie destiné aux régions recevant plus de 2 600 unités thermiques de croissance. Il affiche une excellente productivité en lignes à écartement large ou à écartement étroit et le grain convient pour la cuisson et pour la mise en conserve.

Mots clés: *Phaseolus vulgaris* L., haricot blanc, haricot commun, description de cultivar

The white seeded common bean (*Phaseolus vulgaris* L.) cultivar OAC Dublin, variety (CIFA registration no. 6599) was developed by the Department of Plant Agriculture, University of Guelph, Guelph, Ontario. OAC Dublin has an indeterminate bush plant with erect stem and branches and a long vine. The cultivar can be successfully grown in either traditional wide (70 cm) rows or narrower (<40 cm) rows with direct combine harvesting. It is intended for production in areas of Ontario with more than 2600 crop heat units (CHU).

Pedigree and Breeding Methods

OAC Dublin was derived from the cross OAC Rex/M97I082 made in a growth room in 1999. OAC Rex (Michaels et al. 2006), derived from the cross HR20-728/3/ICA Pijao/PI 440795//Ex Rico23, is resistant to common bacterial blight (caused by *Xanthomonas axonopodis* pv. *phaseoli*). HR20-728 is a selection from the cross Ex Rico 23-Are/Midnight with upright plant architecture and a high podding canopy. PI 440795 is a *Phaseolus acutifolius* plant introduction line, which was used in inter-specific crosses to introgress resistance to common bacterial blight into common bean (Parker 1985). M97I082 was selected from the double-cross OAC Speedvale/WR46//A55/UI114. OAC Speedvale is a navy bean variety. WR46 is a snap bean variety with root rot tolerance. A55 is a black bean line from Cornell University, New York, USA. UI144 is a pinto bean

variety from University of Idaho, USA. The F₁ plants were grown in the field in 1999 at the University of Guelph Elora Research Station, near Elora, Ontario, and all seeds were bulk harvested. The F₂ to F₄ generations were advanced using a modified bulk method, in which a single pod was harvested from each individual and bulked. The F₂ and F₄ generations were grown in a winter nursery in New Zealand. The F₃ generation was grown at the Elora Research Station in the summer of 2000. Single plant selections for early maturity, upright plant architecture, and high pod number were made from F₅ population plots in the field at the Elora Research Station in 2001. The F₅ derived F₆ lines (F_{5:6}) were grown in 2002 at the Elora Research Station in progeny rows. The F_{5:7} selections were grown in preliminary yield trials at the Elora Research station. Selected lines from the preliminary yield trials were tested in advanced yield trials in two locations at the Elora Research Station and in a farmer's field near St. Thomas, Ontario in collaboration with Dr. S. J. Park, AAFC-Harrow. Selection criteria in the preliminary and advanced yield trials were the same agronomic traits as for the F₅ population as well as high seed yield.

Performance

OAC Dublin was entered into the Ontario Pulse Crop Committee's Registration and Performance trials in 2005 and 2006 as OAC 05-2. Registration trials were planted in four locations in 2006 (Kippen, Kemptville, Granton and Elora) and five locations in 2005 (Kippen, St. Thomas, Kemptville, Granton and Elora). During

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2005 and 2006, the line was also grown in seed increase plots at the Elora Research Station to produce F_{5:9} and F_{5:10} generations. One hundred single plants from the F_{5:11} generation, grown in a growth room in the winter of 2006–2007, were rogued for uniformity and trueness to type, and the seed from the remaining plants was bulked individually and planted in plant rows near Twin Falls, Idaho, USA, for breeder seed production in 2007.

Canned bean samples of OAC Dublin grown at four locations were assessed for appearance, flavour and texture by a panel at Agriculture and Agri-Food Canada Greenhouse and Processing Crops Research Centre, Harrow, Ontario. Canned bean texture was mechanically evaluated using the Ottawa texture measurement system for firmness (N mm⁻¹) and plateau force (N) (Voisey 1971) using wire extrusion cells by the Instron texture measurement system model 441 (Instron Corporation, Canton, MA). A visual examination of the degree of packing of canned beans was performed. The washed-drained weight of the cooked samples was determined to quantify percent solid weight of each sample. The hydration coefficient was estimated as the water uptake of 1000 g of dry beans.

In the Ontario White Bean variety registration Trials in 2005 and 2006, OAC Dublin, on average, had the same time to maturity as OAC Thunder (Michaels et al. 1999), but matured 4 and 6 days earlier than the other check cultivars, OAC Gryphon and OAC Rex, respectively (Table 1). Therefore, this variety is appropriate for the short season areas in Ontario with greater than 2600 heat units (Brown and Bootsma 1997). Over 9 location-years, OAC Dublin out-yielded OAC Rex by 11% (411 kg ha⁻¹), OAC Gryphon by 12% (437 kg ha⁻¹) and OAC Thunder by 12% (445 kg ha⁻¹) (Table 1). The cultivar has an average seed weight, similar to OAC Rex. The overall cooking quality of OAC Dublin was rated acceptable by the panellists (Table 2) and is similar to both OAC Thunder and OAC Gryphon.

OAC Dublin is susceptible to common races of anthracnose [caused by *Collectotrichum lindemuthianum* (Sacc. & Mangus)] races 17(alpha) and 23 (delta). The trials were conducted under controlled conditions in

growth chambers at the Greenhouse and Processing Crops Research Centre (GPCRC), Harrow, ON. For each test 10–15 plants were inoculated by brushing both the upper and lower surfaces of the unifoliate leaves with a suspension of 10⁶ *C. lindemuthianum* spores mL⁻¹, in Mathur's medium (0.1% yeast extract, 0.1% Bacto Peptone, 1% sucrose, 0.25% MgSO₄·7H₂O, 0.27% KH₂PO₄, 2% agar supplemented with 25 mg of ampicillin in 1 L of sterile distilled water). Inoculated plants were then placed in a mist chamber with 100% humidity at 23°C for 48 h, and then transferred to a growth cabinet at 23/18°C day/night temperature with a 14-h photoperiod. Disease ratings were performed 5 d after inoculation and were repeated 3 d later.

OAC Dublin is resistant to bean common mosaic virus (BCMV) race 1. The trials were conducted under controlled conditions in growth chambers at the GPCRC, Harrow, ON, using plants with fully expanded unifoliate leaves (approximately 10 d old). Inoculum was maintained by harvesting seeds from infected plants and prepared from infected leaves by grinding them with a mortar and pestle in a 0.01 M phosphate buffer (pH = 7.0) with carborundum powder (Fisher Scientific, Pittsburgh, PA). The upper surfaces of the unifoliate leaves were covered with the race 1 or race 15 inoculums. The leaves were allowed to dry and the plants were misted with water to enhance inoculum absorption. Inoculated plants were maintained in 23/18°C day/night temperature with a 14 h photoperiod. Disease ratings were performed 7–10 d after inoculation.

Although OAC Dublin has OAC Rex in its pedigree it is not resistant to CBB.

Other Characteristics

OAC Dublin has a green hypocotyl in the seedling stage and white flowers, which appear approximately 43 d after planting. At maturity, plants are similar in height to OAC Rex. Pods at maturity are light beige, although slight purple flecking may occur in some environments. Slightly smaller in size than those of OAC Thunder, the leaves of OAC Dublin are similar in size and colour to

Table 1. Average maturity, yield and seed weight of OAC Dublin and three check cultivars in the Ontario Pulse Crop Committee White Bean Variety Registration Trials^z (2005 and 2006)

Cultivar	Yield (kg ha ⁻¹)			Maturity (d)			Seed weight (g 100seed ⁻¹)
	2006	2005	Mean ^y	2006	2005	Mean ^y	
OAC Dublin	4178	4168	4172	97.0	91.8	94	21.8
OAC Gryphon	3492	3929	3735	100.3	96.8	98	21.4
OAC Rex	3766	3758	3761	105.0	96.8	100	21.7
OAC Thunder	3599	3830	3727	96.8	92.8	94	23.3
SE ^x	66.2	67.5	64.8	0.70	0.48	0.56	0.3
No. of trials	4	5	9	4	5	9	9

^zTrials were conducted at St. Thomas, Kippen, Brussels, Elora, Granton, and Kemptville, but not all were used every year.

^yBased on total number of trials.

^xBased on trial means of 27 cultivars.

Table 2. Cooking quality and disease reaction of OAC Dublin and two check cultivars in the Ontario Pulse Crop Committee White Bean Variety Registration Trials^z (2005)

	Cooking quality						Disease reaction			
	Panel ^y	Texture ^x		Packing ^w	Drained wt. ^v	Hydration coefficient ^u	BCMV ^t		Anthracnose ^s	
Cultivar	(1–15)	Plateau force (N)	Firmness (N mm ^{−1})	(1–5)	(%)	(%)	1	15	17	23
OAC Dublin	9.0	291.7	24.4	4.3	69.0	1.83	–	+	+	+
OAC Gryphon	8.9	360.1	28.2	3.2	65.6	1.84	–	–	–	–
OAC Thunder	9.4	338.4	29.4	3.2	65.4	1.82	–	–	+	+
SE ^r	0.11	5.79	0.58	0.10	0.31	0.004				

^zTrials were grown at St. Thomas, Granton, Kippen, Elora.

^yEight panellists evaluated the appearance, flavour and texture of canned beans in tomato sauce using a 1–15 scale; 1, poor; 15, excellent.

^xTexture of washed–drained canned beans measured in Instron Texture measurement system using wire extrusion cells for plateau force (N) and firmness (N mm^{−1}).

^wDegree of clumping of canned beans scored visually from 1 to 5: 1, no clumping; 5, over half clumped.

^vDrained weight: weight of washed–drained canned beans expressed as a percentage of unwashed–undrained weight.

^uSoaked weight/dry weight (determined for 500 g of dry beans soaked for 40 min at 88°C).

^tResistance (–) or susceptibility (+) to bean common mosaic virus (BCMV) races 1 and 15.

^sResistance (–) or susceptibility (+) to Anthracnose races 17 and 23.

^rBased on trial means of 28 cultivars.

those of OAC Rex. Seeds are dull white in colour and similar in size to OAC Rex seeds.

Maintenance and Distribution of Pedigreed Seed

Breeder seed of OAC Dublin is maintained by the Department of Plant Agriculture, University of Guelph, Guelph, ON, Canada N1G 2W1. Pedigreed seed is distributed by R. T. Bolton and Sons, 43234 Winthrop Road, RR#1 Dublin, Ontario, Canada N0K 1E0 and Rosebank Seed Farms Ltd., 7340 Perth Line 24, RR#2 Staffa, Ontario, Canada N0K 1Y0.

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